

CICS 221: INTERNET AND INFORMATION SECURITY
HOME ASSIGNMENT
PROBLEM SHEET

- Q1. Explain why integrity is as important as confidentiality in online banking.
- Q2. State two differences between authentication and identification.
- Q3. Give two examples of security threats that affect availability.
- Q4. Why is hashing passwords better than storing them in plain text?
- Q5. What is the role of salt in password hashing?
- Q6. Mention two weaknesses of password-based authentication.
- Q7. Define malware and give two examples.
- Q8. Why is Kerckhoffs' Principle critical in cryptography?
- Q9. Differentiate between symmetric and asymmetric encryption with one example each.
- Q10. What is a digital signature and what does it guarantee?
- Q11. Why is software complexity a security risk?
- Q12. State two advantages of two-factor authentication.
- Q13. What is the Confused Deputy problem?
- Q14. List two types of firewalls and their operating layers.

Q15. Why is an application proxy firewall more secure than a packet filter?

Q16. What is CAPTCHA and why is it used?

Q17. Define fraud rate and insult rate in biometrics.

Q18. Why is iris recognition considered highly reliable?

Q19. What is anomaly-based IDS and its main advantage?

Q20. Why is key length important in encryption?